

Engine



Special Tool(s)

 ST1341-A	2,200# Floor Crane, Fold Away 300-OTC1819E
 ST3101-A	Bar, Engine Spreader 303-1454
 ST2976A	Eye, Engine Lift 303-1245
 ST1377-A	Lifting Bracket, Engine 303-F047 (014-00073)
 ST1636-A	Retainer, Torque Converter 307-346 (T97T-7902-A) or equivalent

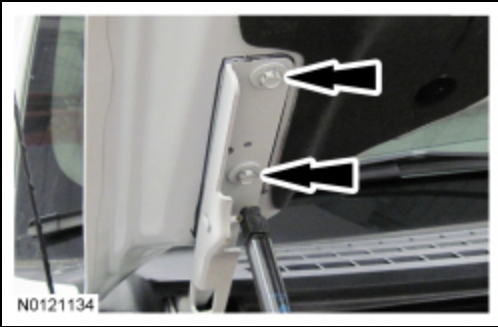
WARNING: Do not smoke, carry lighted tobacco or have an open flame of any type when working on or near any fuel-related component. Highly flammable mixtures are always present and may be ignited. Failure to follow these instructions may result in serious personal injury.

WARNING: Before working on or disconnecting any of the fuel tubes or fuel system components, relieve the fuel system pressure to prevent accidental spraying of fuel. Fuel in the fuel system remains under high pressure, even when the engine is not running. Failure to follow this instruction may result in serious personal injury.

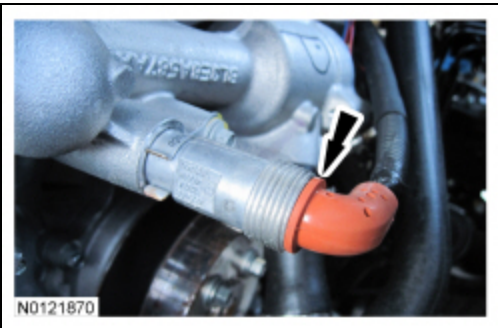
1. With the vehicle in NEUTRAL, position it on a hoist. For additional information, refer to Section 100-02.
2. **NOTE:** Index-mark the hood hinge location to aid in hood installation.

NOTE: RH shown, LH similar.

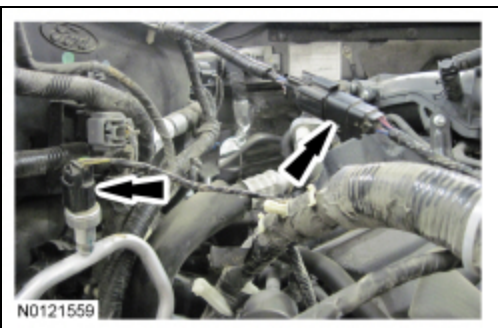
Remove the 4 bolts (2 shown) and the hood.



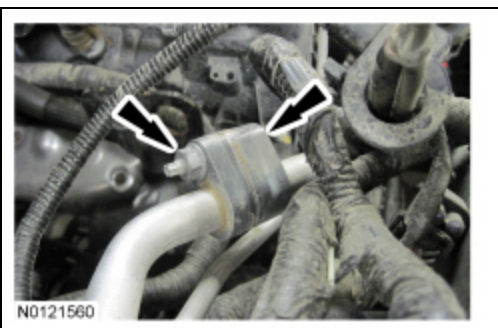
3. Release the fuel system pressure. For additional information, refer to Section 310-00.
4. Disconnect the battery ground cable. For additional information, refer to Section 414-01.
5. If equipped, disconnect the block heater electrical connector at the thermostat housing.
 - Detach the block heater wiring harness retainer from the LH valve cover stud bolt, generator wiring harness and position aside.



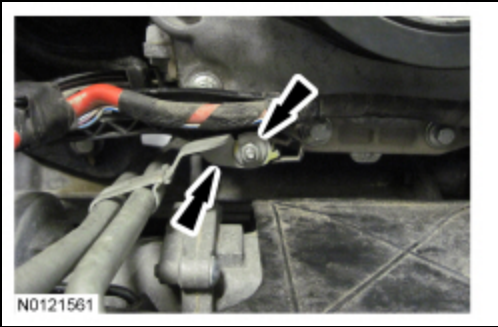
6. Remove the cooling module. For additional information, refer Section 303-03.
7. Remove the Air Cleaner (ACL) upper tray. For additional information, refer to Intake Air System Components — Exploded View, 3.7L in Section 303-12.
8. Disconnect the Battery Junction Box (BJB) wiring harness and the A/C pressure transducer electrical connectors.



9. Remove the nut and disconnect the A/C tube.
 - Discard the O-ring seal and gasket seal.



10. Remove the nut and the transmission cooler line bracket from the engine front cover stud bolt.

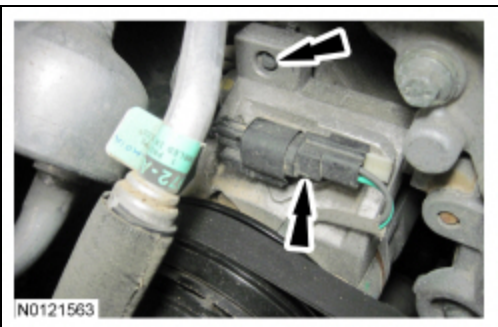


11. Detach the generator wiring harness from the engine front cover.



12. Disconnect the A/C compressor electrical connector.

- Detach the A/C compressor wiring harness from the A/C compressor.



13. Remove the generator. For additional information, refer to Section 414-00.

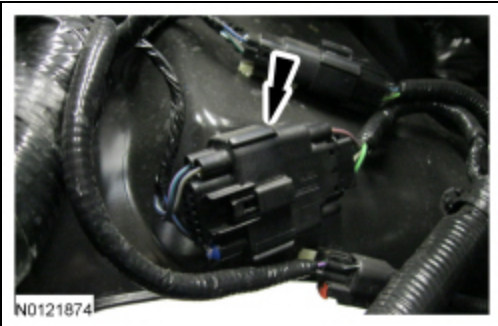
14. Remove the Air Conditioning (A/C) compressor belt. For additional information, refer to Section 303-05.

15. Disconnect the middle PCM electrical connector.

- Detach the wiring harness retainer from the fender splash shield.

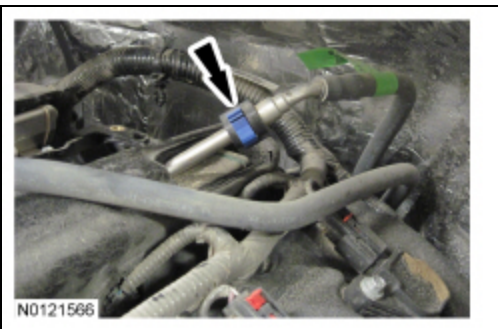


16. Detach and disconnect the engine harness electrical connector.

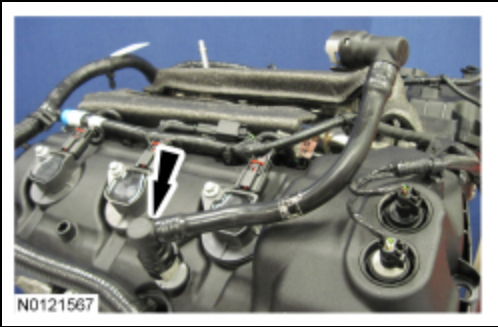


17. Remove the upper intake manifold. For additional information, refer to Upper Intake Manifold in this section.

18. Disconnect the fuel supply tube quick connect coupling from the fuel rail. For additional information, refer to Section 310-00.



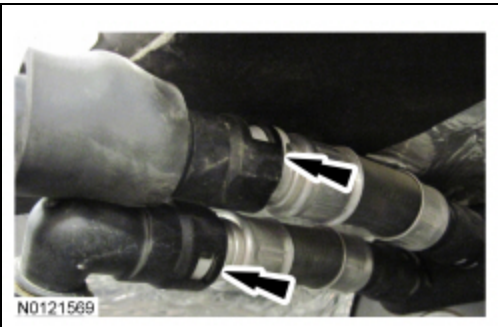
19. Remove the PVC tube quick connect coupling from RH valve cover. For additional information, refer to Section 310-00.



20. Remove the crankcase vent tube quick connect coupling from LH valve cover. For additional information, refer to Section 310-00.



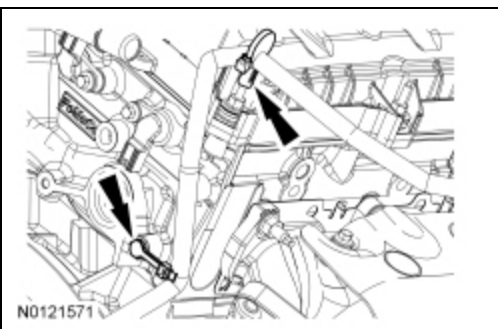
21. Disconnect the 2 heater hose couplings at the rear of the engine.



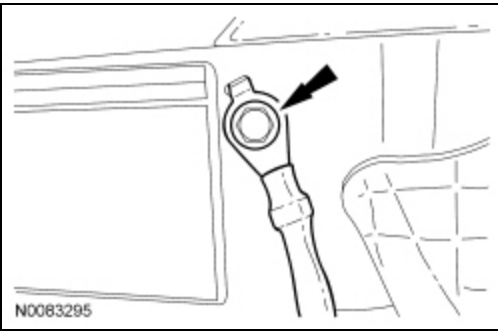
22. Disconnect the 2 coolant hoses from the degas bottle.



23. Detach the wiring harness retainers from the rear of the RH valve cover stud bolt and the rear of the RH cylinder head.



24. Remove the bolt and the ground strap from the cowl.



25. Remove the cowl panel grille. For additional information, refer to Section 501-02.

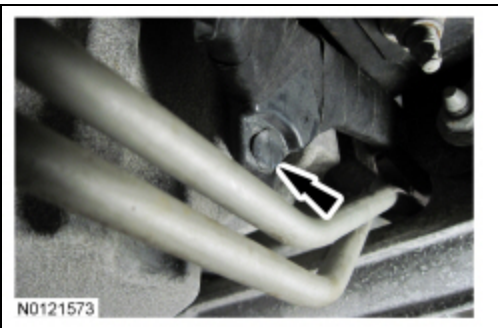
26. Remove the 6 screws and release the 2 cowl panel clips and remove the cowl panel.



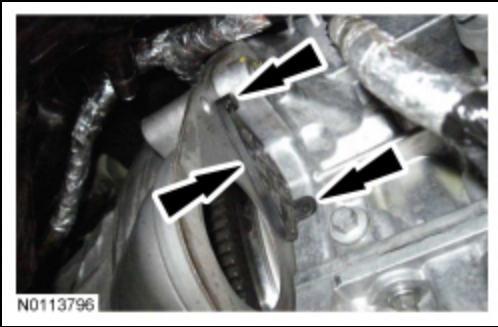
27. If equipped, remove the front driveshaft. For additional information, refer to Section 205-01.

28. Remove the starter motor. For additional information, refer to Section 303-06.

29. Detach the starter motor wiring harness retainer from the oil pan.



30. Remove the 2 pin-type retainers and the access cover.



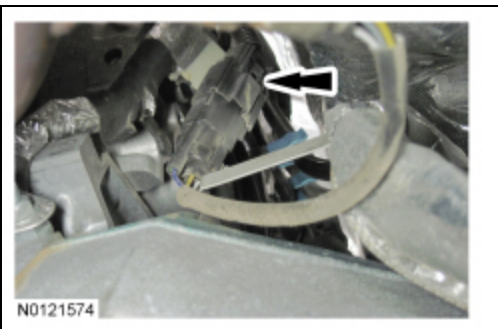
31. **NOTE:** *Index-mark the end of one torque converter stud and the flexplate for installation.*

Remove the 4 torque converter nuts.



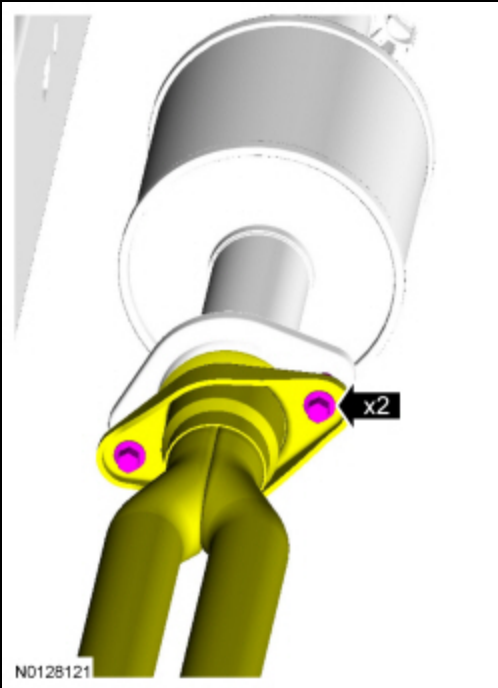
32. **NOTE:** *LH shown, RH similar.*

Disconnect the LH and RH Heated Oxygen Sensor (HO2S) electrical connectors.



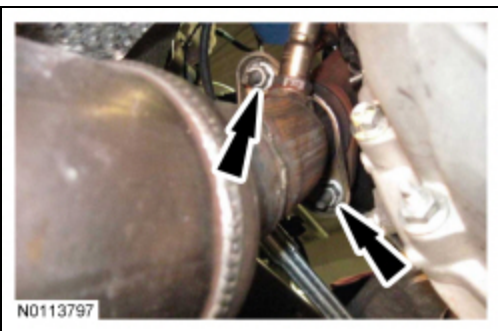
33. Remove the 2 exhaust Y-pipe dual catalytic converter-to-exhaust intermediate pipe bolts and separate.

- Discard the 2 bolts.



34. **NOTE:** *LH shown, RH similar.*

Remove and discard the 2 LH and RH catalytic converter-to-exhaust manifold nuts.



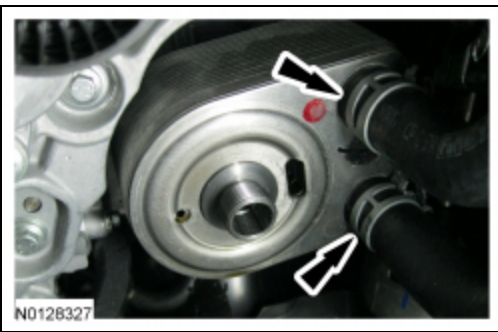
35. Remove the drain plug and drain the engine oil.
- Install the drain plug and tighten to 27 Nm (20 lb-ft).



36. Remove and discard the engine oil filter.



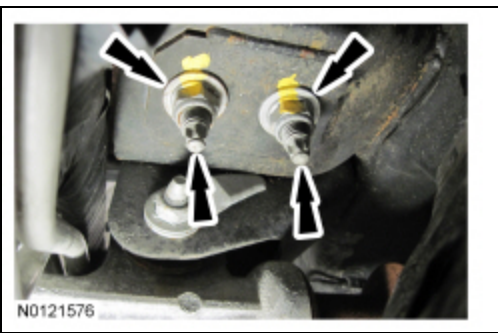
37. If equipped, remove the 2 coolant hoses from the oil cooler.



38. **NOTICE:** Only use hand tools when removing the engine support insulator nuts and studs or damage to the engine support insulator can occur.

NOTE: The 2 engine support insulator studs may come off with the nuts.

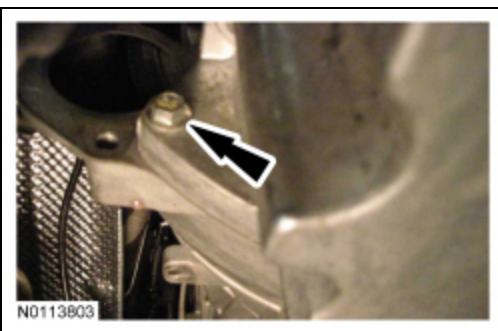
Remove and discard the 2 RH engine support insulator nuts. Remove the engine support insulator studs.



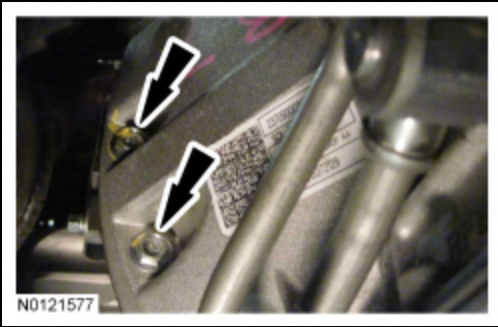
39. Remove the engine-to-bellhousing bolt.



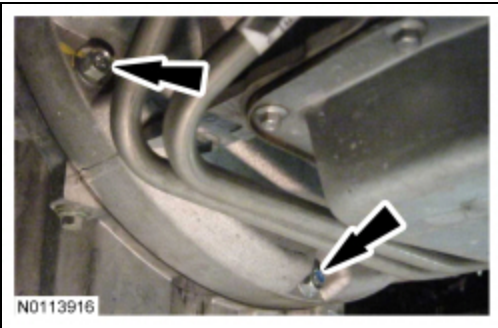
40. Remove the oil pan-to-bellhousing bolt.



41. Remove the 2 LH bellhousing-to-engine bolts.

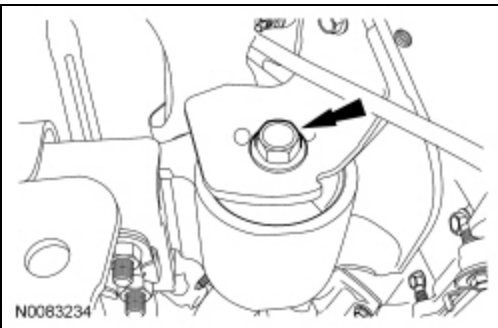


42. Remove the 2 LH bellhousing-to-oil pan bolts.



43. **NOTICE:** Only use hand tools when loosening or tightening the engine support insulator through bolts or damage to the engine support insulator-to-cylinder block bracket can occur.

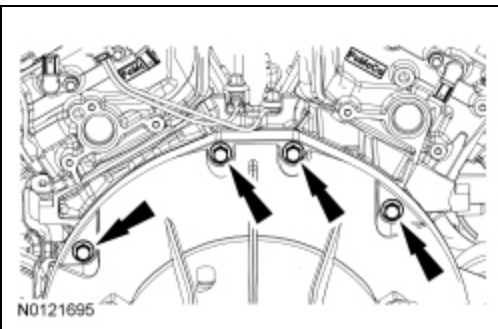
Remove and discard the LH engine mount through bolt.



44. **NOTICE:** Do not support the transmission by the fluid pan, failure to follow this instruction may result in serious damage to the transmission.

Support the bellhousing of the transmission with a suitable floor jack and a block of wood.

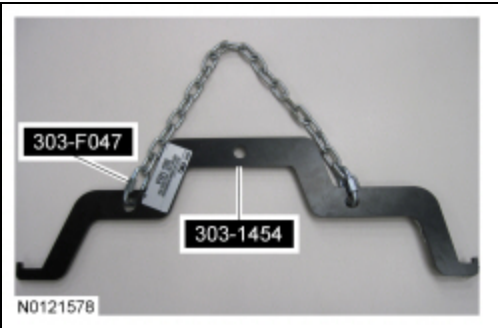
45. Remove the 4 upper bellhousing-to-engine bolts.



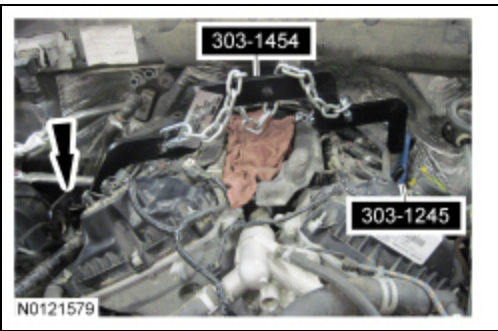
46. Install the LH engine lift eye.



47. Using part of Engine Lifting Bracket and a chain, assemble the Engine Spreader Bar as shown.

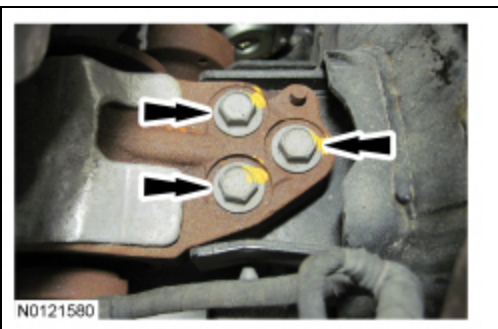


48. Install the Engine Spreader Bar assembly into the Engine Lift Eyes.



49. **NOTE:** After removing the bolts leave the LH engine support insulator in place.

Remove the 3 bolts for the LH engine support insulator.

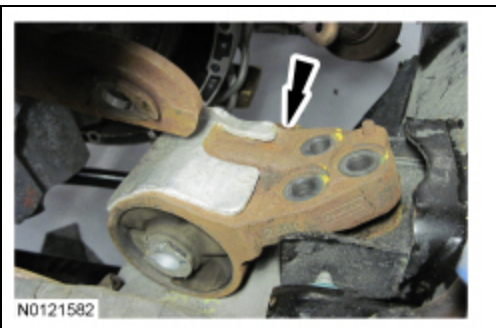


50. **NOTE:** The use of a ratchet strap may be needed to level the engine.

Using the Heavy Duty Floor Crane and Engine Spreader Bar assembly, raise the engine for clearance to remove the LH engine support insulator.



51. Remove the LH engine support insulator.



52. Detach the generator wiring harness retainer from the lower LH side of the engine block.



53. Using the Heavy Duty Floor Crane and Engine Spreader Bar assembly, remove the engine.



54. Install the Torque Converter Retainer.

